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Task 10:

Centralized Security Architecture

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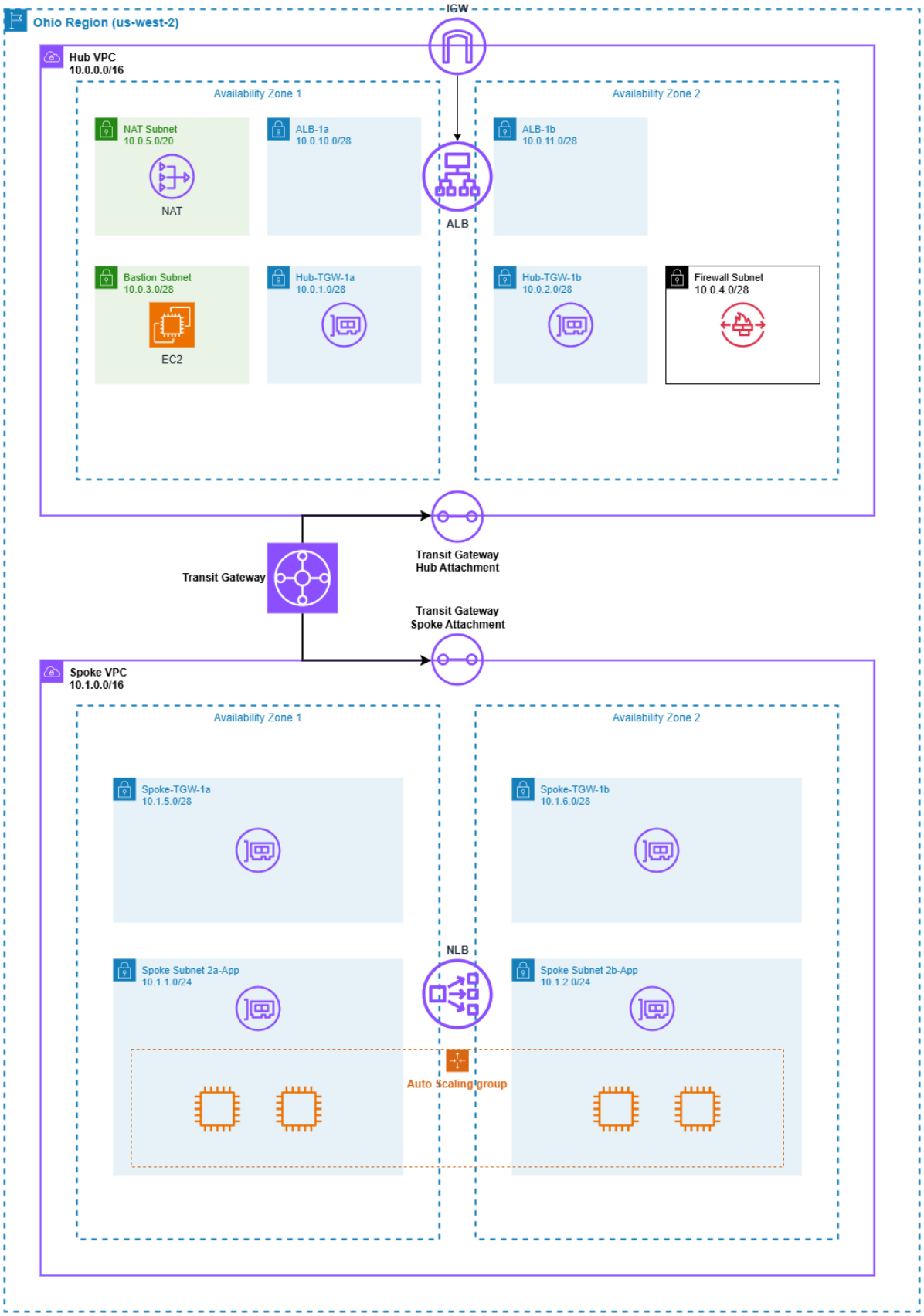
Task Documentation: Centralized Security and Ingress Architecture

1. Service Overview: Core Components of the Architecture

This architecture implements a Hub-and-Spoke network topology designed for centralized traffic inspection and routing. By leveraging specific AWS networking services, the design ensures that all traffic—whether entering from the internet, leaving to the internet, or moving between VPCs—passes through a central inspection point.

- **AWS Transit Gateway (TGW):** Acts as a cloud router, connecting the Hub-VPC and multiple Spoke-VPCs. It simplifies network topology by removing the need for complex VPC peering meshes.
- **AWS Network Firewall:** A managed firewall service deployed in the Hub-VPC. It provides stateful inspection, intrusion prevention, and web filtering to protect the network.
- **Application Load Balancer (ALB):** Deployed in the Hub-VPC to handle incoming internet traffic. It routes requests through the firewall and Transit Gateway to backend resources in private Spoke-VPCs.
- **Gateway Load Balancer Endpoints (GWLBE):** Used in route tables to direct traffic to the Network Firewall transparently, ensuring traffic flows are symmetric and inspected.
- **Zonal NAT Gateway:** A network address translation service placed in a specific availability zone to ensure strict control over egress routing paths, allowing outbound traffic to be inspected by the firewall before reaching the internet.


Ohio Region (us-west-2)
IGW
Hub VPC
10.0.0.0/16
Availability Zone 1
Availability Zone 2
NAT Subnet
10.0.5.0/20
NAT
ALB-1a
10.0.10.0/28
ALB
ALB-1b
10.0.11.0/28
Bastion Subnet
10.0.3.0/28
EC2
Hub-TGW-1a
10.0.1.0/28
Hub-TGW-1b
10.0.2.0/28
Firewall Subnet
10.0.4.0/28
Transit Gateway
Transit Gateway
Hub Attachment
Transit Gateway
Spoke Attachment
Spoke VPC
10.1.0.0/16
Availability Zone 1
Availability Zone 2
Spoke-TGW-1a
10.1.5.0/28
Spoke-TGW-1b
10.1.6.0/28
NLB
Spoke Subnet 2a-App
10.1.1.0/24
Spoke Subnet 2b-App
10.1.2.0/24
Auto Scaling group

2. Project Context & Objectives

This section outlines the business need for this centralized architecture and the roles involved in managing the infrastructure.

The Scenario

An enterprise organization requires a scalable and secure network architecture. They have application workloads running in isolated "Spoke" VPCs that contain private resources (Auto Scaling Groups). However, for security compliance, these applications cannot be exposed directly to the internet.

Instead, all ingress (incoming) and egress (outgoing) traffic must be routed through a central "Hub" VPC. This Hub acts as the perimeter security layer. The challenge is to route traffic from an internet-facing ALB in the Hub, through a firewall, over a Transit Gateway, to the application in the Spoke, and ensure the return traffic follows the exact same path (symmetry) to avoid connection drops.

Key Stakeholders

- **Network Architects:** Responsible for configuring the Transit Gateway, Route Tables, and ensuring connectivity between the Hub and Spokes.
- **Security Engineers:** Responsible for deploying the AWS Network Firewall, defining stateful/stateless rule groups, and monitoring traffic for threats.
- **DevOps/Application Teams:** Responsible for deploying the web applications and Auto Scaling Groups in the Spoke-VPCs, relying on the Hub for internet connectivity.
- **Compliance Officers:** Require assurance that all traffic flows are logged and inspected by a central security appliance.

Primary Objectives

1. **Centralized Ingress Security:** Route all incoming internet traffic through a Hub ALB and Force it through the Network Firewall before it reaches the application.
2. **Secure Egress Filtering:** Ensure private instances in Spoke VPCs can access the internet (for updates) only by passing through the Hub Firewall and a Zonal NAT Gateway.
3. **Routing Symmetry:** Implement advanced routing logic (using specific route tables and TGW Appliance Mode) to ensure request and response packets follow the same path through the firewall, preventing asymmetric routing drops.
4. **Scalability:** Allow for the addition of new Spoke VPCs without modifying the central security infrastructure.

5.

3. Implementation Guide: Step-by-Step Task Execution

Prerequisites: Network Planning

- Before beginning the deployment, strict IP addressing (CIDR) planning is required to prevent overlapping ranges between the Hub, Spokes, and TGW attachments.
- **Hub-VPC:** 10.0.0.0/16
- **Spoke-VPC:** 10.1.0.0/16

Hello! I am your AWS Solution Architect, and I will guide you through this complex centralized security architecture. This design requires strict adherence to routing dependencies, so we will break this down into logical phases to prevent configuration errors.

We will start with the foundation: **VPC and Subnet Creation**.

Phase 1: VPC and Subnet Setup

We need to create the network topology exactly as specified to ensure the IP routing we configure later works correctly.

Step 1.1: Create the Hub-VPC

1. Navigate to the **VPC Console** > **Your VPCs**.
2. Click **Create VPC**.
3. **Name tag:** Hub-VPC
4. **IPv4 CIDR block:** 10.0.0.0/16
5. **Tenancy:** Default.
6. Click **Create VPC**.

Step 1.2: Create Hub-VPC Subnets

Go to **Subnets** > **Create subnet**. Select Hub-VPC and create the following sequentially (you can add multiple new subnets in the create screen to do this in one batch):

Name	Availability Zone	CIDR Block	Purpose
Firewall-Subnet	**us-east-1a** (or your preferred AZ 1)	10.0.4.0/28	Network Firewall Endpoint
Hub-TGW-1a	**us-east-1a**	10.0.1.0/28	TGW Attachment
Hub-TGW-1b	**us-east-1b** (or your preferred AZ 2)	10.0.2.0/28	TGW Attachment
ALB-1a	**us-east-1a**	10.0.10.0/28	Load Balancer
ALB-1b	**us-east-1b**	10.0.11.0/28	Load Balancer
Bastion-Subnet	**us-east-1b**	10.0.3.0/28	Management Host
NAT-Subnet	**us-east-1a**	10.0.5.0/28	NAT Gateway

Note: I have assigned 10.0.5.0/28 to the NAT Subnet as requested.

Step 1.3: Create the Spoke-VPC

1. Go back to **Your VPCs** > **Create VPC**.
2. **Name tag:** Spoke-VPC
3. **IPv4 CIDR block:** 10.1.0.0/16[1]
4. Click **Create VPC**.

Step 1.4: Create Spoke-VPC Subnets

Go to **Subnets** > **Create subnet**. Select Spoke-VPC and create:

Name	Availability Zone	CIDR Block
Spoke-TGW-Att-1a	**us-east-1a**	10.1.5.0/28
Spoke-TGW-Att-1b	**us-east-1b**	10.1.6.0/28
Spoke-App1-Subnet-1a	**us-east-1a**	10.1.1.0/24
Spoke-App2-Subnet-1b	**us-east-1b**	10.1.2.0/24

Phase 2: Core Connectivity Infrastructure

- We need these components operational to obtain their IDs (Gateway IDs, Attachment IDs) for the routing configuration in the next phase.

Step 2.1: Create and Attach Internet Gateway (IGW)

1. Navigate to **VPC Console** > **Internet gateways**.
2. Click **Create internet gateway**.
3. **Name:** Hub-IGW.
4. Click **Create internet gateway**.
5. Click **Actions** > **Attach to VPC**.
6. Select **Hub-VPC** and confirm.

Step 2.2: Create NAT Gateway

- This is required for the Spoke-VPC to access the internet (Egress) through the Hub.

1. Navigate to **NAT gateways** > **Create NAT gateway**.
2. **Name:** Hub-NAT-GW.
3. **Subnet:** Select NAT-Subnet (10.0.5.0/28) in the **Hub-VPC**.
4. **Availability mode**: Zonal
5. **Connectivity type:** Public.
6. **Elastic IP allocation:** Click **Allocate Elastic IP**.
7. Click **Create NAT gateway**.
    - *Note: It may take a minute to become "Available," but we can proceed.*

Step 2.3: Create Transit Gateway (TGW)

1. Navigate to **Transit gateways** > **Transit gateways**.
2. Click **Create transit gateway**.
3. **Name tag:** Main-TGW.
4. **Amazon side ASN:** Keep default (e.g., 64512).
5. **Auto accept shared attachments:** Enable (optional, but convenient).
6. **Default route table association/propagation:** Enable both.
7. Click **Create transit gateway**.

Step 2.4: Create Transit Gateway Attachments

- We must explicitly tell the TGW which subnets to use.
- **Attachment A: Hub-VPC**

1. Go to **Transit gateway attachments** > **Create transit gateway attachment**.
2. **Name tag:** Hub-TGW-Attachment.
3. **Transit gateway ID:** Select Main-TGW.
4. **Attachment type:** VPC.
5. **VPC ID:** Select Hub-VPC.
6. **Subnet IDs:** Select **only** the dedicated TGW subnets:

    - Hub-TGW-1a (10.0.1.0/28)
    - Hub-TGW-1b (10.0.2.0/28)
7. Click **Create transit gateway attachment**.

- **Attachment B: Spoke-VPC**

1. Click **Create transit gateway attachment** again.
2. **Name tag:** Spoke-TGW-Attachment.
3. **Transit gateway ID:** Select Main-TGW.
4. **Attachment type:** VPC.
5. **VPC ID:** Select Spoke-VPC.
6. **Subnet IDs:** Select **only** the dedicated TGW subnets:
    - Spoke-TGW-Att-1a (10.1.5.0/28)
    - Spoke-TGW-Att-1b (10.1.6.0/28)
7. Click **Create transit gateway attachment**.

Phase 3: Network Firewall Configuration

- We need to create the logic (Rule Groups & Policy) before creating the physical Firewall resource.

Step 3.1: Create Rule Groups

- Navigate to the **VPC Console** > **Network Firewall** > **Network Firewall rule groups**.
- **A. Create Stateless Rule Group (Pass-Through)**

1. Click **Create rule group**.
2. **Name:** Stateless-Allow-All.
3. **Type:** Stateless.
4. **Capacity:** 100.
5. **Rule group format:** Standard.
6. Click **Create**.
7. **Add Rule:** On the configuration page, under **Rules**, click **Add rule**.
    - **Protocol:** All.
    - **Source/Destination:** Any (0.0.0.0/0).
    - **Action: Pass** (This is critical; "Pass" allows it to move to Stateful processing or next actions).
    - Click **Add**.

- **B. Create Stateful Rule Group (For Testing)**

1. Go back and click **Create rule group**.
2. **Name:** Stateful-Test-Rules.
3. **Type:** Stateful.
4. **Capacity:** 100.
5. **Format: Suricata compatible IPS rules**. (This gives us flexibility for the specific DENY test later).
6. **Rule string:** Enter a generic pass rule for now (or leave blank if allowed, but a pass rule is safer). Enter: pass ip any any -> any any (msg:"Allow all"; sid:1000001; rev:1;)
7. Click **Create stateful rule group**.

Step 3.2: Create Firewall Policy

- Navigate to **Network Firewall policies**.

1. Click **Create firewall policy**.
2. **Name:** Hub-Security-Policy.
3. **Stream exception policy:** Continue.
4. **Stateless default actions:**
    - Operation order: Forward to stateful rule groups.
    - (This ensures if traffic doesn't match the stateless "Pass" rule—though it should—it goes to stateful inspection).
5. **Stateless rule groups:** Click **Add rule groups** > Select Stateless-Allow-All.
6. **Stateful rule groups:** Click **Add rule groups** > Select Stateful-Test-Rules.
7. Click **Create firewall policy**.

Step 3.3: Deploy the Network Firewall

- Navigate to **Firewalls**.

1. Click **Create firewall**.
2. **Name:** Hub-Network-Firewall.
3. **VPC:** Select Hub-VPC.
4. **Firewall subnet:**
    - Select Availability Zone **us-east-1a** (The one containing 10.0.4.0/28).
    - Select the **Firewall-Subnet**.
5. **Firewall policy:** Select Hub-Security-Policy.
6. Click **Create firewall**.

Phase 4: Route Table Configuration

- Navigate to **VPC Console** > **Route tables**. We will create and configure 5 distinct route tables.

Step 4.1: Hub - ALB Route Table (Public + Inspection)

- This table ensures the ALB can reach the internet and forces traffic destined for the Spoke-VPC through the Firewall.

1. Click **Create route table**. Name: Hub-ALB-RT. VPC: Hub-VPC. Create.
2. **Edit routes** > **Add route**:
   - 0.0.0.0/0 → **Internet Gateway** (Hub-IGW).
   - 10.1.0.0/16 (Spoke CIDR) → **Gateway Load Balancer Endpoint** (Select the **Firewall Endpoint ID** you copied earlier, usually labeled vpce-...).
   - Click **Save changes**.
3. **Subnet associations** > **Edit subnet associations**:
   - Select ALB-1a and ALB-1b.
   - Click **Save associations**.

Step 4.2: Hub - Firewall Route Table

- This table tells the Firewall where to send traffic after inspection: to the TGW (if going to Spoke) or to the NAT (if going to the Internet).

1. Click **Create route table**. Name: Hub-Firewall-RT. VPC: Hub-VPC. Create.
2. **Edit routes**:
   - 0.0.0.0/0 → **NAT Gateway** (Hub-NAT-GW).
   - 10.1.0.0/16 → **Transit Gateway** (Main-TGW).
   - Click **Save changes**.
3. **Subnet associations**:
   - Select Firewall-Subnet.
   - Click **Save associations**.

Step 4.3: Hub - TGW Route Table

- This table ensures that traffic coming *back* from the Spoke (or entering the Hub via TGW) is sent to the Firewall for inspection before going anywhere else.

1. Click **Create route table**. Name: Hub-TGW-RT. VPC: Hub-VPC. Create.
2. **Edit routes**:
   - 0.0.0.0/0 → **Gateway Load Balancer Endpoint** (The **Firewall Endpoint ID** vpce-...).
   - Click **Save changes**.
3. **Subnet associations**:
   - Select Hub-TGW-1a and Hub-TGW-1b.
   - Click **Save associations**.

Step 4.4: Hub - Public/Bastion Route Table

- This handles the Bastion host and the NAT Gateway itself (which needs a direct path to the internet).

1. Click **Create route table**. Name: Hub-Public-RT. VPC: Hub-VPC. Create.
2. **Edit routes**:
   - 0.0.0.0/0 → **Internet Gateway** (Hub-IGW).
   - 10.1.0.0/16 → **Transit Gateway** (Main-TGW) — *Required so the Bastion can reach the Spoke.*
   - Click **Save changes**.
3. **Subnet associations**:
   - Select Bastion-Subnet and NAT-Subnet.
   - Click **Save associations**.

Step 4.5: Spoke - Main Route Table

- The Spoke-VPC is simple: everything non-local goes to the Transit Gateway.

1. Click **Create route table**. Name: Spoke-Main-RT. VPC: Spoke-VPC. Create.
2. **Edit routes**:
   - 0.0.0.0/0 → **Transit Gateway** (Main-TGW).
   - Click **Save changes**.
3. **Subnet associations**:
   - Select **ALL 4** Spoke subnets (Spoke-TGW-Att-1a/b, Spoke-App1/App2).
   - Click **Save associations**.

Phase 5: Security Groups & Compute Resources

Step 5.1: Create Security Groups

- Navigate to **VPC Console** > **Security Groups**.
- **A. Hub-ALB-SG (For the Load Balancer)**

1. Click **Create security group**.
2. **Name:** Hub-ALB-SG.
3. **VPC:** Select Hub-VPC.
4. **Inbound Rules:**
   - Type: HTTP | Source: Anywhere-IPv4 (0.0.0.0/0).
5. Click **Create**.

- **B. Spoke-Web-SG (For the Web Servers)**

1. Click **Create security group**.
2. **Name:** Spoke-Web-SG.
3. **VPC:** Select Spoke-VPC.
4. **Inbound Rules:**
   - Type: HTTP | Source: Custom (10.0.0.0/16) — *Allows traffic from Hub ALB & specific Hub subnets.*
   - Type: SSH | Source: Custom (10.0.0.0/16) — *Allows SSH from the Bastion host.*
   - Type: ICMP (All ICMP - IPv4) | Source: Custom (10.0.0.0/16) — *Allows Ping verification.*
5. Click **Create**.

- **C. Hub-Bastion-SG (For Management)**

1. Click **Create security group**.
2. **Name:** Hub-Bastion-SG.
3. **VPC:** Select Hub-VPC.
4. **Inbound Rules:**
   - Type: SSH | Source: **My IP** (Select from dropdown) — *Strictly limits access to you.*
5. Click **Create**.

Step 5.2: Launch Spoke Web Servers (Auto Scaling Group)

- We need instances running a web server to test the connectivity.

1. Navigate to **EC2 Console** > **Launch Templates** > **Create launch template**.
2. **Name:** Spoke-Web-Template.
3. **AMI:** Amazon Linux 2023 (or Amazon Linux 2).
4. **Instance type:** t2.micro or t3.micro.
5. **Key pair:** Select an existing key (or create one named vockey).
6. **Network settings:**
   - **Security groups:** Select Spoke-Web-SG. (Note: You might need to refresh the dropdown or verify the VPC is Spoke-VPC if it's not showing).

7. **Advanced details** > **User data** (Scroll to bottom):
   - Paste the following script to create the "Hello" page:

     code Bash

     - downloadcontent_copy
     - expand_less

```
    #!/bin/bash
yum update -y
yum install -y httpd
systemctl start httpd
systemctl enable httpd
echo "<h1>Hello from Spoke VPC</h1>" > /var/www/html/index.html
```

8. Click **Create launch template**.

- Now, create the **Auto Scaling Group**:

1. Navigate to **Auto Scaling Groups** > **Create Auto Scaling group**.
2. **Name:** Spoke-ASG.
3. **Launch Template:** Select Spoke-Web-Template. Click **Next**.
4. **VPC:** Select Spoke-VPC.
5. **Subnets:** Select Spoke-App1-Subnet-1a and Spoke-App2-Subnet-1b. Click **Next**.
6. **Load balancing:** Skip (We are using a Hub ALB, not attaching here). Click **Next**.
7. **Group size:** Desired: 2, Minimum: 2, Maximum: 2.
8. Click **Next** through the options and click **Create Auto Scaling group**.

Step 5.3: Launch Bastion Host

1. Navigate to **Instances** > **Launch instances**.
2. **Name:** Hub-Bastion.
3. **AMI:** Amazon Linux 2023.
4. **Instance type:** t2.micro.
5. **Key pair:** Select your key.
6. **Network settings:**
   - **VPC:** Hub-VPC.
   - **Subnet:** Bastion-Subnet.
   - **Auto-assign Public IP: Enable**.
   - **Security group:** Select Hub-Bastion-SG.
7. Click **Launch instance**.

Phase 6: Application Load Balancer (ALB) Configuration

Step 6.1: Create Target Group

1. Navigate to **EC2 Console** > **Target Groups** > **Create target group**.
2. **Choose a target type:** Select **IP addresses**. (This is critical because the instances are in a different VPC).
3. **Target group name:** Spoke-Web-TG.
4. **Protocol:** HTTP | **Port:** 80.
5. **IP address type:** IPv4.
6. **VPC:** Select Hub-VPC. (The Load Balancer lives here).
7. **Health check protocol:** HTTP | **Path:** / (Default).
8. Click **Next**.

- **Register Targets:**

1. **Network:** Select **Other private IP addresses**.
2. **IPv4 address:** Enter the **Private IP** of your first Spoke instance. Click **Include as pending below**.
3. Repeat for the second Spoke instance IP.
4. You should see both IPs listed in the "Targets" review area.
5. Click **Create target group**.

Step 6.2: Create the Application Load Balancer

1. Navigate to **Load Balancers** > **Create load balancer**.
2. **Load balancer type:** Application Load Balancer (ALB) -> Click **Create**.
3. **Name:** Hub-Public-ALB.
4. **Scheme: Internet-facing**.
5. **IP address type:** IPv4.
6. **Network mapping:**
   - **VPC:** Hub-VPC.
   - **Mappings:**
     - Check **us-east-1a** -> Select subnet ALB-1a.
     - Check **us-east-1b** -> Select subnet ALB-1b.
7. **Security groups:**
   - Remove default.
   - Select Hub-ALB-SG (The one we created earlier).
8. **Listeners and routing:**
   - **Protocol:** HTTP | **Port:** 80.
   - **Default action:** Select Spoke-Web-TG.
9. Click **Create load balancer**.

Phase 7: Testing and Verification

- Perform these tests in order.

Test 1: Functional Routing (The "Happy Path")

- This confirms that: Internet → ALB → Firewall → TGW → Spoke Web Server.

1. Copy the **DNS name** of your Hub-Public-ALB.
2. Paste it into a new browser tab and hit Enter.
3. **Result:** You should see the white page with the text: **"Hello from Spoke VPC"**.
   - *If you see this, your Route Tables, TGW Attachments, and Security Groups are perfectly configured.*

Test 2: Firewall Inspection (The "Block" Test)

- We will now prove the Firewall is active by intentionally blocking this specific traffic.

1. Navigate to **VPC Console** > **Network Firewall** > **Network Firewall rule groups**.
2. Select Stateful-Test-Rules.
3. Click **Edit rules**.
4. In the rule entry box, **add a new line** (below the existing "pass" rule) with this **DROP** command:

   code Text

   - downloadcontent_copy
   - expand_less

   ```
   drop tcp 10.0.0.0/16 any -> 10.1.0.0/16 80 (msg:"Block Hub ALB to Spoke"; sid:1000002; rev:1;)
   ```

   - *(Note: This creates a signature ID 1000002 that drops TCP traffic from the Hub CIDR to the Spoke CIDR on port 80).*

5. Click **Save rule group**.
6. **Verify:** Go back to your browser tab with the ALB and refresh the page.
   - **Result:** The page should **spin endlessly** (timeout) or eventually show **502 Bad Gateway**. This confirms the Firewall successfully intercepted and dropped the packet.
7. **Restore:** Edit the rule group again and **remove** the drop line you just added. Save the rule group. Refresh your browser to confirm connectivity is restored.

Test 3: Internal Connectivity & Egress

- We will verify that the Spoke instances can reach the internet (e.g., for updates) via the Hub NAT Gateway.

1. **Connect to the Bastion:**

- Open your terminal/Putty.
- SSH into the Bastion Host public IP: ssh -i key.pem ec2-user@<Bastion-Public-IP>

2. **Jump to the Spoke Instance:**
   - Since your private key isn't on the Bastion, create a temporary one:
     - Run nano key.pem on the Bastion.
     - Paste the content of your private key file.
     - Save (Ctrl+O, Enter) and Exit (Ctrl+X).
     - Run chmod 400 key.pem.
   - SSH to one of the **Spoke Private IPs** (e.g., 10.1.1.x):
     ssh -i key.pem ec2-user@10.1.1.x
3. **Test Egress (Ping 8.8.8.8):**
   - Run: ping 8.8.8.8
   - **Result:** You should see successful replies (64 bytes from 8.8.8.8...).
   - *This proves: Spoke → TGW → Firewall → NAT GW → Internet works.*

Conclusion

- Congratulations! You have successfully built a **Centralized Inspection Architecture** on AWS.
- **Summary of what you built:**

1. **Ingress Security:** Internet traffic hits the Hub ALB, gets routed into the Network Firewall for inspection, traverses the Transit Gateway, and finally hits the application in the Spoke VPC.
2. **Egress Security:** Servers in the Spoke VPC send traffic to the Hub via TGW, which passes through the Firewall before exiting via the NAT Gateway.
3. **Network Isolation:** The Spoke VPC has no direct Internet Gateway; it is fully dependent on the Hub for secure connectivity.

- **Clean Up (Optional):**

If you are done with this lab, remember to delete resources in reverse order to avoid costs:

1. Delete Spoke ASG & Launch Template.
2. Delete ALB & Target Group.
3. Delete Instances (Bastion).
4. Delete Network Firewall, Policy, and Rule Groups.
5. Delete NAT Gateway & Release Elastic IP.
6. Delete Transit Gateway Attachments -> Delete Transit Gateway.
7. Delete Subnets -> Delete VPCs.